Class 9 Mathematics — Exercise 5.1

Basic Concept

(1) Linear Equation

An equation in which the **power (exponent) of the variable is always 1** is called a linear equation. The variable is the **unknown**.

$$2x + 3 = 0$$

A linear equation is also called an **equation of degree one**.

(2) Types

(i) Linear equation in one variable: $2x + 3 = 0$

(ii) Linear equation in two variables: $2x + y = 3$

(3) Point to Ponder — Checking a Solution

While checking a solution, we write a **question mark over the equal sign** ($\overset{?}{=}$), because we are not yet sure that both sides are equal.

Example: $-7x + 24 = 3$

$$-7x + 24 - 24 = 3 - 24$$

$$-7x = -21$$

$$x = \frac{-21}{-7} \quad \Rightarrow \quad x = 3$$

Check:

$$-7(3) + 24 \overset{?}{=} 3$$

$$-21 + 24 \overset{?}{=} 3$$

$$3 = 3$$

So $x = 3$ is the required root.

Method used in every question: Solve for the variable → Check (L.H.S = R.H.S) → write the Solution Set (S.S).

Question 1

$$5x - 2 - x = 4 - 3x - 27$$

Solution:

$$5x - x + 3x = 4 - 27 + 2$$

$$7x = -21$$

$$x = \frac{-21}{7}$$

$$\boxed{x = -3}$$

Check:

$$5(-3) - 2 - (-3) = 4 - 3(-3) - 27$$

$$-15 - 2 + 3 = 4 + 9 - 27$$

$$-17 + 3 = 13 - 27$$

$$-14 = -14$$

L.H.S = R.H.S

$$\boxed{S.S = \{-3\}}$$

Question 2

$$4a - 3(5a - 14) = 5(7 + a) - 9$$

Solution:

$$4a - 15a + 42 = 35 + 5a - 9$$

$$4a - 15a - 5a = 35 - 9 - 42$$

$$-16a = -16$$

$$a = \frac{16}{16}$$

$$\boxed{a = 1}$$

Check:

$$4(1) - 3[5(1) - 14] = 5[7 + (1)] - 9$$

$$4 - 3(5 - 14) = 5(7 + 1) - 9$$

$$4 - 3(-9) = 5(8) - 9$$

$$4 + 27 = 40 - 9$$

$$31 = 31$$

L.H.S = R.H.S

$$\boxed{S.S = \{1\}}$$

Question 3

$$7(2-5x)+27=18x-3(8-4x)$$

Solution:

$$14-35x+27=18x-24+12x$$
$$-35x-18x-12x=-24-14-27$$
$$-65x=-65$$
$$x=\frac{65}{65}$$
$$\boxed{x=1}$$

Check:

$$7[2-5(1)]+27=18(1)-3[8-4(1)]$$
$$7(-3)+27=18-3(4)$$
$$-21+27=18-12$$
$$6=6$$

L.H.S = R.H.S

$$\boxed{S.S=\{1\}}$$

Question 4

$$\frac{5x}{4}+\frac{1}{2}=0$$

Solution:

LCM of 4 and 2:

$$\begin{array}{r|l} 2 & 4,\ \ 2 \\ 2 & 2,\ \ 1 \\ \hline & 1,\ \ 1 \end{array} \qquad \text{LCM} = 2\times 2 = 4$$

Multiply by 4 on both sides:

$$4\left(\frac{5x}{4}\right)+4\left(\frac{1}{2}\right)=4\times 0$$
$$5x+2=0$$
$$5x=-2$$
$$\boxed{x=-\frac{2}{5}}$$

Check:

$$\frac{5\left(-\frac{2}{5}\right)}{4}+\frac{1}{2}=0$$

$$-\frac{2}{4}+\frac{1}{2}=0$$
$$-\frac{1}{2}+\frac{1}{2}=0$$
$$0=0$$

L.H.S = R.H.S

$$\boxed{S.S=\left\{-\frac{2}{5}\right\}}$$

Question 5

$$\frac{x-2}{2}+\frac{x+10}{9}=5$$

Solution:

LCM of 2 and 9:

2	2, 9
3	1, 9
3	1, 3
	1, 1

$$\text{LCM}=2\times3\times3=18$$

Multiply by 18 on both sides:

$$9(x-2)+2(x+10)=18\times5$$
$$9x-18+2x+20=90$$
$$11x+2=90$$
$$11x=88$$
$$x=\frac{88}{11}$$
$$\boxed{x=8}$$

Check:

$$\frac{8-2}{2}+\frac{8+10}{9}=5$$
$$\frac{6}{2}+\frac{18}{9}=5$$
$$3+2=5$$
$$5=5$$

L.H.S = R.H.S

$$\boxed{S.S=\{8\}}$$

Question 6

$$\frac{4(x+2)}{3} - \frac{6(x-7)}{7} = 12$$

Solution:

LCM of 3 and 7 = 21. Multiply by 21 on both sides:

$$7 \times 4(x+2) - 3 \times 6(x-7) = 21 \times 12$$
$$28(x+2) - 18(x-7) = 252$$
$$28x + 56 - 18x + 126 = 252$$
$$28x - 18x = 252 - 56 - 126$$
$$10x = 70$$
$$x = \frac{70}{10}$$
$$\boxed{x = 7}$$

Check:

$$\frac{4(7+2)}{3} - \frac{6(7-7)}{7} = 12$$
$$\frac{4(9)}{3} - \frac{6(0)}{7} = 12$$
$$12 - 0 = 12$$
$$12 = 12$$

L.H.S = R.H.S

$$\boxed{S.S = \{7\}}$$

Question 7

$$\frac{x}{2} + \frac{x}{3} - \frac{x}{4} + \frac{x}{5} = 7\frac{5}{6}$$

Solution:

$$\frac{x}{2} + \frac{x}{3} - \frac{x}{4} + \frac{x}{5} = \frac{47}{6}$$

LCM of 2, 3, 4, 5, 6:

2	2,	3,	4,	5,	6
2	1,	3,	2,	5,	3
3	1,	3,	1,	5,	3
5	1,	1,	1,	5,	1
	1,	1,	1,	1,	1

$$\text{LCM} = 2 \times 2 \times 3 \times 5 = 60$$

Multiply by 60 on both sides:

$$30x + 20x - 15x + 12x = 10 \times 47$$
$$47x = 470$$
$$x = \frac{470}{47}$$
$$\boxed{x = 10}$$

Check:

$$\frac{10}{2} + \frac{10}{3} - \frac{10}{4} + \frac{10}{5} = \frac{47}{6}$$
$$\frac{300 + 200 - 150 + 120}{60} = \frac{47}{6}$$
$$\frac{470}{60} = \frac{47}{6}$$
$$\frac{47}{6} = \frac{47}{6}$$

L.H.S = R.H.S

$$\boxed{S.S = \{10\}}$$

Question 8

$$\frac{y+1}{3} + \frac{y+1}{2} = 2 - \frac{y+3}{2}$$

Solution:

$$\frac{y+1}{3} + \frac{y+1}{2} = \frac{2}{1} - \frac{y+3}{2}$$

LCM of 3, 2, 1 = 6. Multiply by 6 on both sides:

$$2(y+1) + 3(y+1) = 6(2) - 3(y+3)$$
$$2y + 2 + 3y + 3 = 12 - 3y - 9$$
$$2y + 3y + 3y = 12 - 9 - 2 - 3$$
$$8y = -2$$
$$y = -\frac{2}{8}$$
$$\boxed{y = -\frac{1}{4}}$$

Check:

$$\frac{-\frac{1}{4} + 1}{3} + \frac{-\frac{1}{4} + 1}{2} = 2 - \frac{-\frac{1}{4} + 3}{2}$$
$$\frac{\frac{-1+4}{4}}{3} + \frac{\frac{-1+4}{4}}{2} = 2 - \frac{\frac{-1+12}{4}}{2}$$

$$\frac{3}{12}+\frac{3}{8}=\frac{2}{1}-\frac{11}{8}$$

Note: $\dfrac{3/4}{3}=\dfrac{3}{4}\div\dfrac{3}{1}=\dfrac{3}{4}\times\dfrac{1}{3}=\dfrac{3}{12}$

$$\frac{1}{4}+\frac{3}{8}=\frac{2}{1}-\frac{11}{8}$$
$$\frac{2+3}{8}=\frac{16-11}{8}$$
$$\frac{5}{8}=\frac{5}{8}$$

L.H.S = R.H.S

$$\boxed{S.S=\left\{-\frac{1}{4}\right\}}$$

Question 9

$$\frac{1}{5}(x-8)+\frac{4+x}{7}=7-\frac{23-x}{5}$$

Solution:

LCM of 5 and 7 = 35. Multiply by 35 on both sides:

$$7(x-8)+5(4+x)=35(7)-7(23-x)$$
$$7x-56+20+5x=245-161+7x$$
$$12x-36=84+7x$$
$$12x-7x=84+36$$
$$5x=120$$
$$x=\frac{120}{5}$$
$$\boxed{x=24}$$

Check:

$$\frac{1}{5}(24-8)+\frac{4+24}{7}=7-\frac{23-24}{5}$$
$$\frac{16}{5}+\frac{28}{7}=7-\frac{-1}{5}$$
$$\frac{16}{5}+4=7+\frac{1}{5}$$
$$\frac{16+20}{5}=\frac{35+1}{5}$$
$$\frac{36}{5}=\frac{36}{5}$$

L.H.S = R.H.S

$$\boxed{S.S=\{24\}}$$

Question 10

$$\frac{1}{2y} - \frac{1}{6} = \frac{1}{4y} - 1 - \frac{1}{y}\ ; \qquad y \neq 0$$

Solution:

LCM of $2y, 6, 4y, y$:

2	2, 6, 4
2	1, 3, 2
3	1, 3, 1
	1, 1, 1

$$\text{LCM} = 2 \times 2 \times 3 \times y = 12y$$

Multiply by $12y$ on both sides:

$$6 - 2y = 3 - 12y - 12$$
$$-2y + 12y = 3 - 12 - 6$$
$$10y = -15$$
$$y = -\frac{15}{10}$$
$$\boxed{y = -\frac{3}{2}}$$

Check:

$$\frac{1}{2\left(-\frac{3}{2}\right)} - \frac{1}{6} = \frac{1}{4\left(-\frac{3}{2}\right)} - 1 - \frac{1}{-\frac{3}{2}}$$
$$-\frac{1}{3} - \frac{1}{6} = -\frac{1}{6} - \frac{1}{1} + \frac{2}{3}$$
$$\frac{-2-1}{6} = \frac{-1-6+4}{6}$$
$$-\frac{3}{6} = -\frac{3}{6}$$
$$-\frac{1}{2} = -\frac{1}{2}$$

L.H.S = R.H.S

$$\boxed{S.S = \left\{-\frac{3}{2}\right\}}$$

Question 11

$$4 - 0.3(1 - x) = 7$$

Solution:

$$4 - 0.3 + 0.3x = 7$$
$$0.3x = 7 - 4 + 0.3$$

$$0.3x = 3.3$$
$$x = \frac{3.3}{0.3}$$
$$\boxed{x = 11}$$

Check:

$$4 - 0.3(1 - x) = 7$$
$$4 - 0.3(1 - 11) = 7$$
$$4 - 0.3(-10) = 7$$
$$4 + 3 = 7$$
$$7 = 7$$

L.H.S = R.H.S

$$\boxed{S.S = \{11\}}$$

Question 12

$$0.5x = 6.3 - 0.2x$$

Solution:

$$0.5x + 0.2x = 6.3$$
$$0.7x = 6.3$$
$$x = \frac{6.3}{0.7} = \frac{63/10}{7/10} = \frac{63}{7}$$
$$\boxed{x = 9}$$

Check:

$$0.5(9) = 6.3 - 0.2(9)$$
$$4.5 = 6.3 - 1.8$$
$$4.5 = 4.5$$

L.H.S = R.H.S

$$\boxed{S.S = \{9\}}$$

Question 13

$$1.3x - 0.2 = 0.3x - 1.5$$

Solution:

$$1.3x - 0.3x = -1.5 + 0.2$$

$$1x = -1.3$$

$$\boxed{x = -1.3}$$

Check:

$$1.3x - 0.2 = 0.3x - 1.5$$

$$1.3(-1.3) - 0.2 = 0.3(-1.3) - 1.5$$

$$-1.69 - 0.2 = -0.39 - 1.5$$

$$-1.89 = -1.89$$

L.H.S = R.H.S

$$\boxed{S.S = \{-1.3\}}$$